General Dessert #10: RFI to Consumer Electronics, Grounding, and Bonding

Diagnosing RFI by its audible signature, bypass capacitors and ferrite chokes, common-mode current, ground loops, lightning protection, resonant grounds, and chassis bonding.

Group G4C · 12 pool questions · Self-study reference · Practice exams at eham.net/exams

Key Facts Reference

Diagnosing RFI by Symptom

Symptom or scenario	Cause / answer	Q
Audio device hears RFI from an SSB transmitter – what does it sound like?	**Distorted speech.** SSB demodulates poorly through a junction; voice is recognizable but garbled.	G4C03
Audio device hears RFI from a CW transmitter	**On-and-off humming or clicking.** CW keying creates pulses, not a tone.	G4C04
Cause of interference spanning a wide range of frequencies	**Arcing at a poor electrical connection.** Broadband noise source.	G4C02

Fixing RFI

Technique	Where / why it helps	Q
Reducing RFI into audio circuits	**Bypass capacitor** across the audio input to shunt RF to ground.	G4C01
Reducing RFI from common-mode current on an audio cable	**Place a ferrite choke on the cable.** Adds series impedance to common-mode current only.	G4C08

Grounding and Bonding

Question	Answer	Q
Cause of high voltages producing RF burns	**The ground wire has high impedance on that frequency** – the “ground” is not at RF ground.	G4C05
Effect of a resonant ground connection	**High RF voltages on the enclosures of station equipment.**	G4C06
Soldered joints in lightning protection ground	Should not be used – **a soldered joint will likely be destroyed by the heat of a lightning strike.** Use mechanical/exothermic.	G4C07
Minimizing ground loops	**Bond equipment enclosures together** (single-point or short low-impedance bonding).	G4C09
Symptom of a ground loop in audio	**Reports of “hum” on your transmitted signal.**	G4C10
Minimizing RF “hot spots” in the station	**Bonding all equipment enclosures together.**	G4C11
Why ground metal enclosures of station equipment	**To ensure hazardous voltages cannot appear on the chassis** (AC safety, primary reason).	G4C12

Dessert 10 Question Bank (G4C)

G4C01. Which of the following might be useful in reducing RF interference to audio frequency circuits?
A. Bypass inductor
B. Bypass capacitor
C. Forward-biased diode
D. Reverse-biased diode

G4C02. Which of the following could be a cause of interference covering a wide range of frequencies?
A. Not using a balun or line isolator to feed balanced antennas
B. Lack of rectification of the transmitter's signal in power conductors
C. Arcing at a poor electrical connection
D. Using a balun to feed an unbalanced antenna

G4C03. What sound is heard from an audio device experiencing RF interference from a single sideband phone transmitter?
A. A steady hum whenever the transmitter is on the air
B. On-and-off humming or clicking
C. Distorted speech
D. Clearly audible speech

G4C04. What sound is heard from an audio device experiencing RF interference from a CW transmitter?
A. On-and-off humming or clicking
B. A CW signal at a nearly pure audio frequency
C. A chirpy CW signal
D. Severely distorted audio

G4C05. What is a possible cause of high voltages that produce RF burns?
A. Flat braid rather than round wire has been used for the ground wire
B. Insulated wire has been used for the ground wire
C. The ground rod is resonant
D. The ground wire has high impedance on that frequency

G4C06. What is a possible effect of a resonant ground connection?
A. Overheating of ground straps
B. Corrosion of the ground rod
C. High RF voltages on the enclosures of station equipment
D. A ground loop

G4C07. Why should soldered joints not be used in lightning protection ground connections?
A. A soldered joint will likely be destroyed by the heat of a lightning strike
B. Solder flux will prevent a low conductivity connection
C. Solder has too high a dielectric constant to provide adequate lightning protection
D. All these choices are correct

G4C08. Which of the following would reduce RF interference caused by common-mode current on an audio cable?
A. Place a ferrite choke on the cable
B. Connect the center conductor to the shield of all cables to short circuit the RFI signal
C. Ground the center conductor of the audio cable causing the interference
D. Add an additional insulating jacket to the cable

G4C09. How can the effects of ground loops be minimized?
A. Connect all ground conductors in series
B. Connect the AC neutral conductor to the ground wire
C. Avoid using lock washers and star washers when making ground connections
D. Bond equipment enclosures together

G4C10. What could be a symptom caused by a ground loop in your station's audio connections?
A. You receive reports of "hum" on your station's transmitted signal
B. The SWR reading for one or more antennas is suddenly very high
C. An item of station equipment starts to draw excessive amounts of current
D. You receive reports of harmonic interference from your station

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G4C11. What technique helps to minimize RF "hot spots" in an amateur station?
A. Building all equipment in a metal enclosure
B. Using surge suppressor power outlets
C. Bonding all equipment enclosures together
D. Placing low-pass filters on all feed lines

G4C12. Why must all metal enclosures of station equipment be grounded?
A. It prevents a blown fuse in the event of an internal short circuit
B. It prevents signal overload
C. It ensures that the neutral wire is grounded
D. It ensures that hazardous voltages cannot appear on the chassis

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Answer Key

G4C01: B	G4C02: C	G4C03: C	G4C04: A
G4C05: D	G4C06: C	G4C07: A	G4C08: A
G4C09: D	G4C10: A	G4C11: C	G4C12: D